

Network Bottleneck Audit & Dispatch Optimization Strategy

Executive Summary

An in-depth analysis of the historical delivery network identified the facilities and routing patterns responsible for the majority of SLA breaches and delayed shipments. The study shows that a small number of highly connected hubs create cascading delays across the entire network when congested.

By shifting from a reactive dispatch process to a graph-driven, network-aware strategy, operations can significantly reduce delays, optimize truck allocation, and lower transportation costs.

1. Top Systemic Bottlenecks

The analysis identified five major hubs contributing the highest number of SLA breaches:

- **Gurgaon Bilaspur HB (Haryana):** 1,681 breaches
- **Bhiwandi Mankoli HB (Maharashtra):** 1,403 breaches
- **Bangalore Nelmngla H (Karnataka):** 1,295 breaches
- **Chandigarh Mehmdpur H (Punjab):** 735 breaches
- **Pune Tathawde H (Maharashtra):** 710 breaches

Among these, Gurgaon, Bhiwandi, and Bangalore emerged as the most critical chokepoints due to their high structural importance in the network. Congestion at these facilities impacts a large portion of national freight movement.

To address this, these hubs should be prioritized for operational upgrades such as:

- Increased sorting capacity
- Automated cross-docking systems
- Dedicated overflow handling
- Daily shipment volume caps during peak periods

2. Corridor-Specific Routing Interventions

Certain point-to-point corridors consistently experience high SLA breach volumes and require immediate routing interventions.

Bangalore Intra-City Corridor (Nelmngla → Bomsndra)

This corridor alone contributes more than 250 chronic SLA breaches.

Recommended Action:

Introduce a parallel routing lane or bypass sorting mechanism so that freight moving toward the Bomsndra zone avoids the primary national sorting floor, reducing internal congestion.

Pune → Bhiwandi Corridor

This corridor generates over 100 recurring SLA breaches due to excessive dependence on Carting during high-traffic periods.

Recommended Action:

Shift peak-hour operations on this route from Carting to dedicated FTL dispatches to reduce transit variability and improve delivery reliability.

3. Financial & Operational Impact

Focusing interventions on just the top three hubs—Gurgaon, Bhiwandi, and Bangalore—can produce a major network-wide impact because of their high connectivity and operational influence.

Expected Improvements

- **9.18% reduction in total SLA breaches** across the network
- **2.18 Million in recovered revenue-at-risk** through lower customer penalties, fewer refunds, and reduced expedited shipping costs

This demonstrates that targeted infrastructure and routing improvements can create disproportionate operational gains.

4. Dispatch Optimization Strategy (FTL vs Carting)

The study also evaluated historical dispatch decisions across more than 26,000 trips to determine whether FTL or Carting was the more cost-effective option under different network conditions.

Excessive FTL Usage

The analysis identified **13,664 trips** where premium FTL trucks were used on low-risk corridors that could reliably meet SLA requirements through standard Carting.

Recommendation:

Downgrade these routes to Carting to eliminate unnecessary premium transportation costs.

Under-Protected High-Risk Routes

The model identified **92 high-risk trips** where Carting was used despite originating from severe chokepoints.

Recommendation:

Upgrade these routes to FTL to avoid predictable SLA failures and downstream penalty costs.

Final Outcome

Integrating graph intelligence and risk-aware dispatch optimization into operational workflows is projected to generate approximately **6.07 Million in total cost savings** while simultaneously improving SLA compliance and network stability.

